INOUT

HACKS 2026

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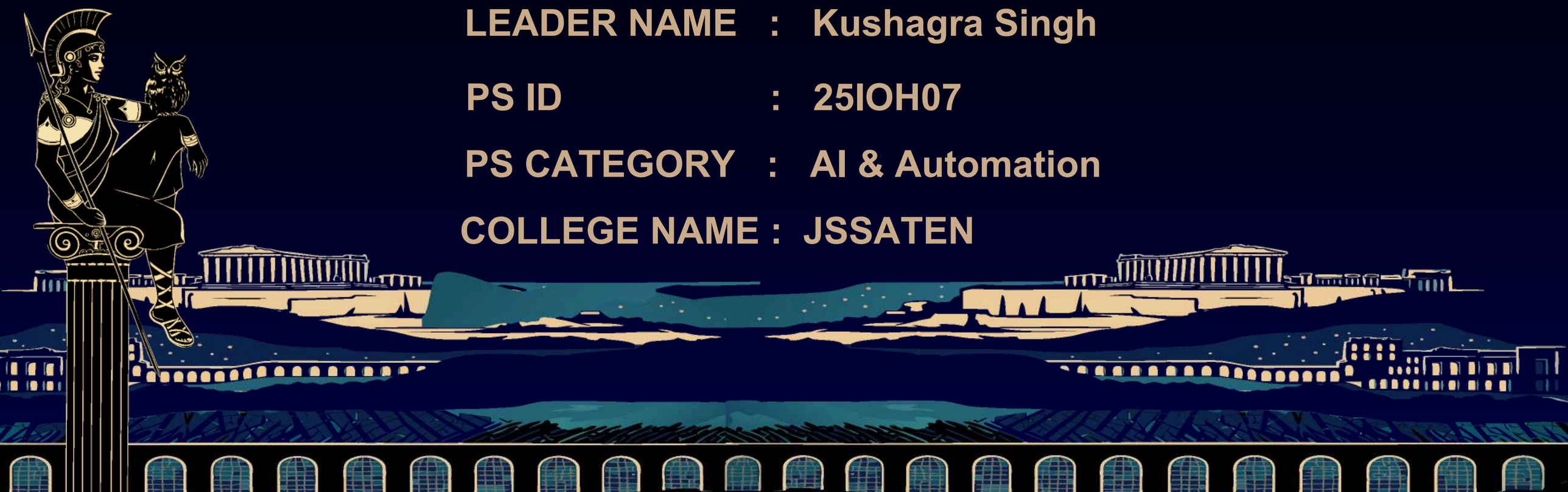
TEAM DETAILS

TEAM NAME	:	HackStreet Boys
LEADER NAME	:	Kushagra Singh
PS ID	:	25IOH07
PS CATEGORY	:	AI & Automation
COLLEGE NAME	:	JSSATEN



The Problem: Dark Patterns Proliferate Unchecked

What Are Dark Patterns?

Manipulative UI/UX designs that trick users into actions they didn't intend—hidden costs, forced continuity, disguised ads, and privacy traps embedded in interfaces.

Current State

Manual audits are time-consuming, expensive, and require legal expertise. Teams lack tools to proactively identify compliance risks before enforcement actions.

IMPACT AND SOLUTION

Transform compliance from reactive legal burden to proactive design advantage. Reduce risk, build trust, and accelerate time-to-market with confident, ethical design choices.

Autonomous Detection

Computer vision and NLP scan websites and apps without human intervention

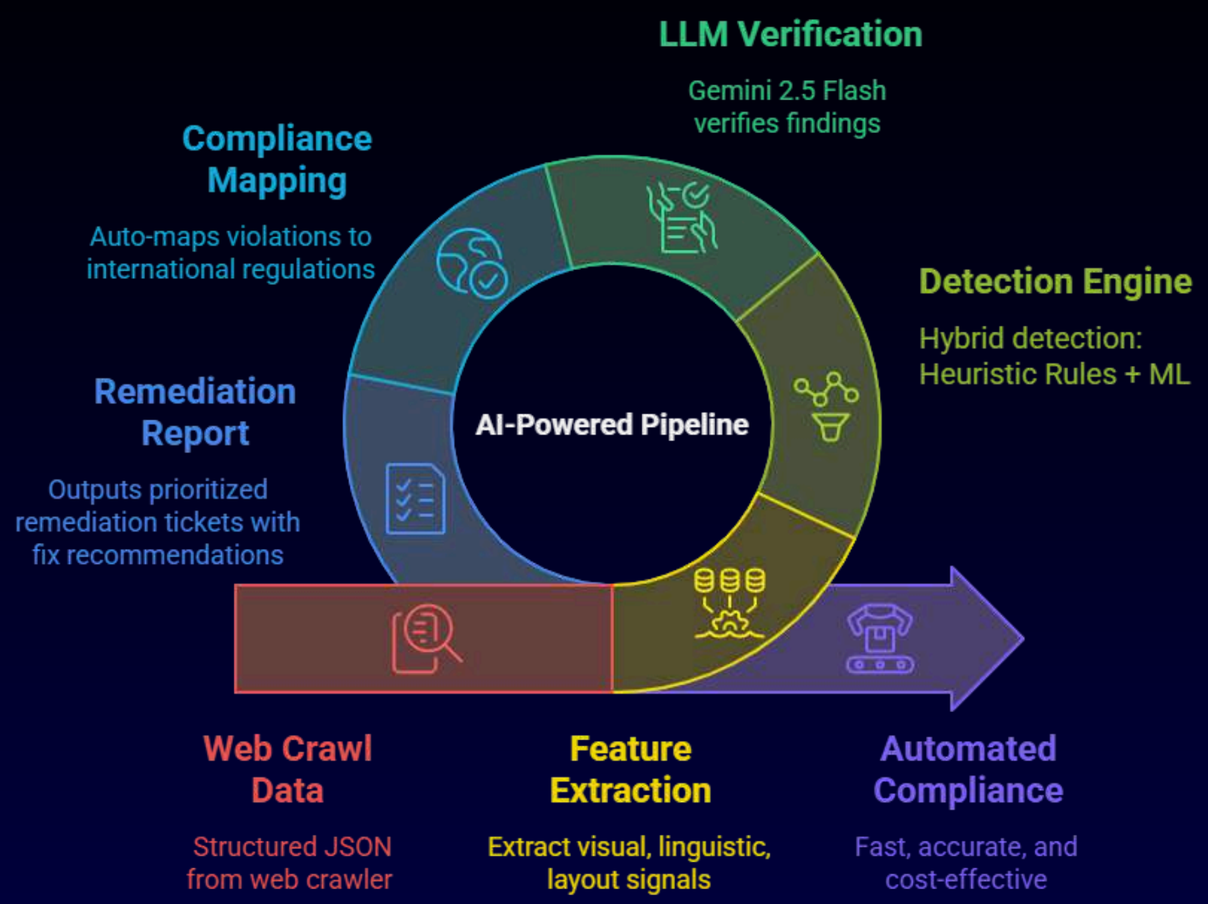
Regulatory Scoring

Map findings to DPDP Act, EU AI Act, and Consumer Protection guidelines

Actionable Roadmap

Generate prioritized remediation plan product teams can execute independently

Architecture (6-Stage Pipeline)



Our Approach — Multi-Modal Detection

CV Agent Visual tricks	NLP Agent Manipulative language	Spatial Agent Layout deception
Image similarity matching	Text classification	Color contrast checks
element size checks	urgency/scarcity detection	button size comparison
pre-checked inputs	guilt-trip patterns	popup/overlay detection
hidden element detection	tone comparison	buried text detection

Why multi-modal?

Dark patterns exploit visuals, language, and layout simultaneously — single-modality tools miss cross-cutting patterns.

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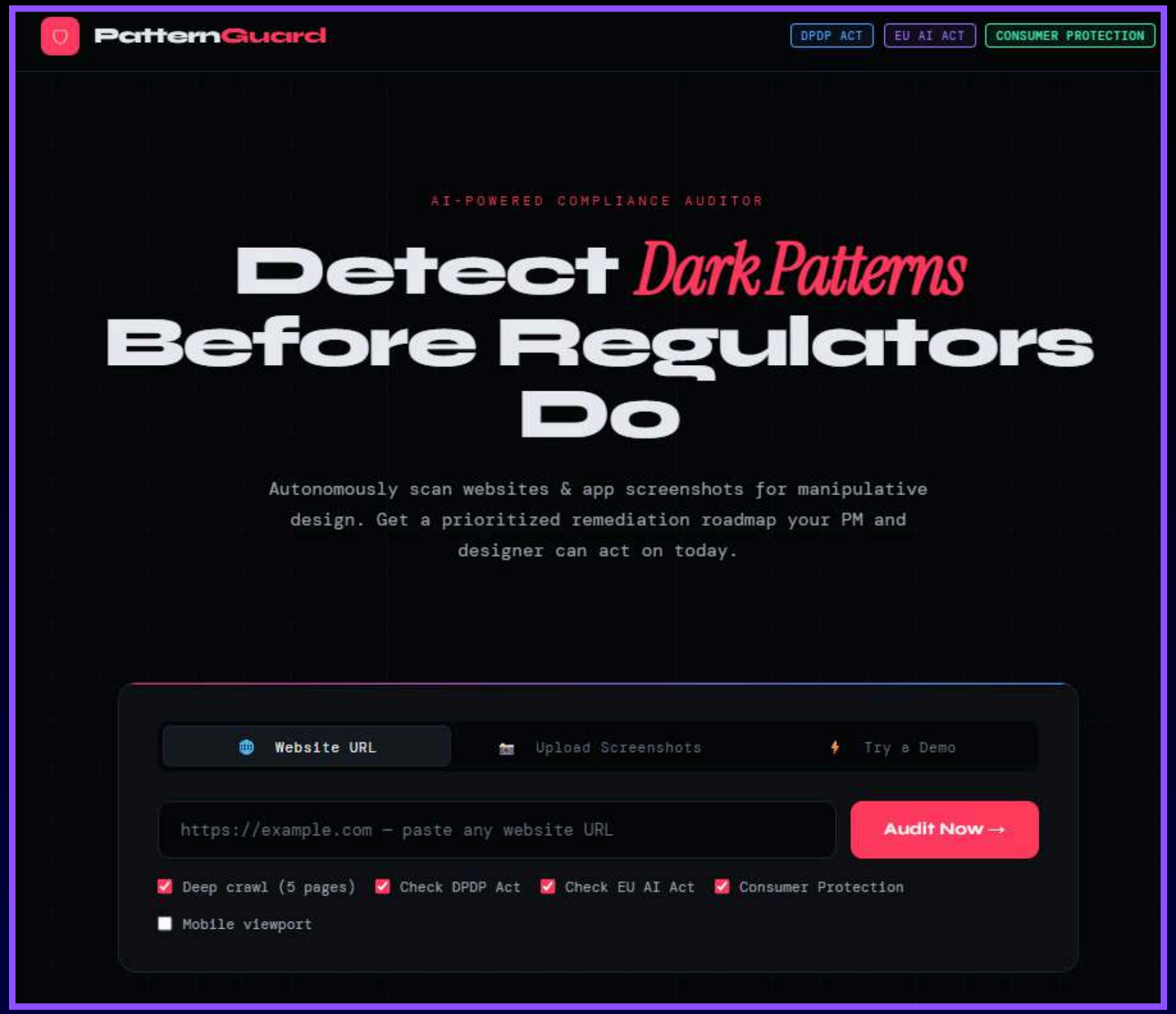
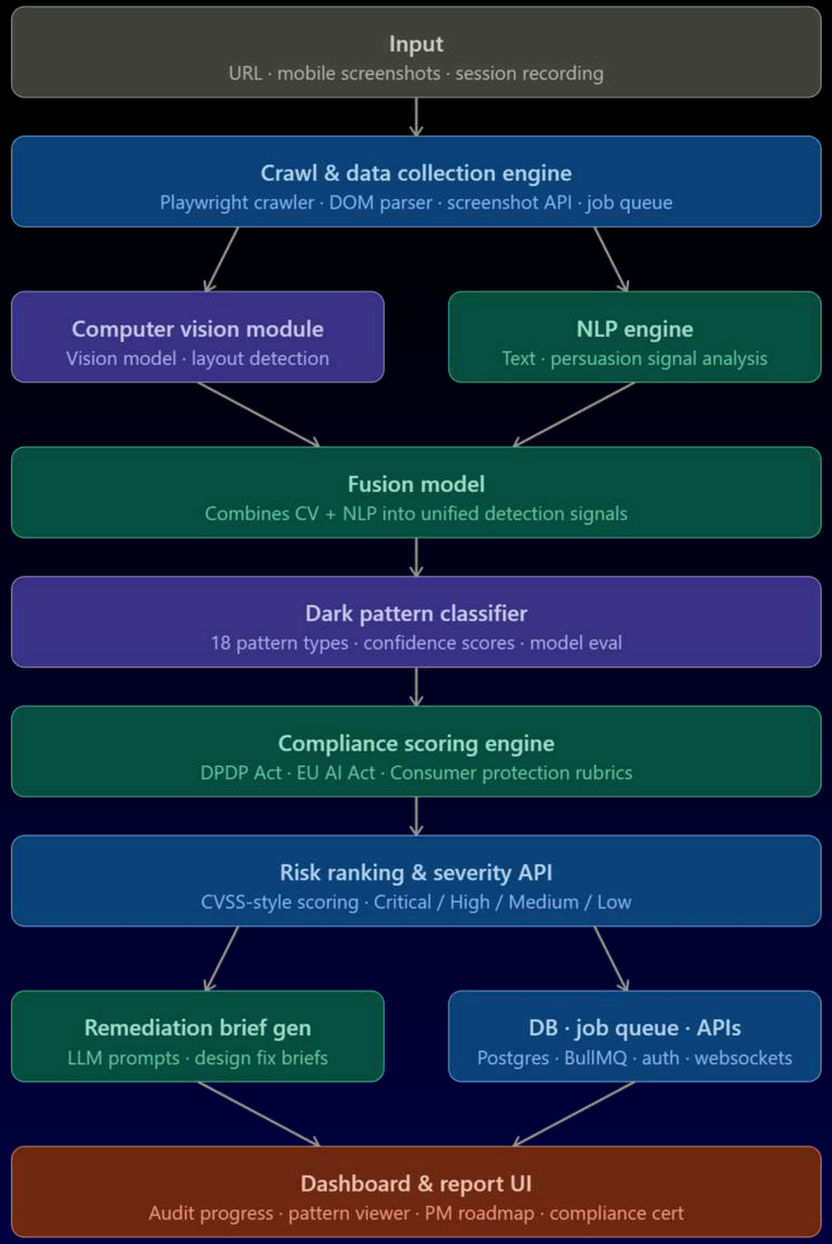
PROTOTYPE





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Market Opportunity

Surging Legal Penalties Skyrocketing global fines are creating urgent demand for automated compliance tool.

Disrupting Expensive Audits We replace slow, costly legal consultants with instant AI-driven compliance resolution.

Empowering Web Designers Translates complex legal regulations into immediate, actionable design fixes.

Massive Untapped Market 200M active websites globally, continuous compliance monitoring is a multi-billion dollar opportunity.

Scalability Model

Freemium Acquisition

Offer a free website vulnerability scan to drive viral traffic, then monetize the AI assistant for step-by-step fixes.

Niche Expansion

Dominate high-fine, high-risk industries like e-commerce and healthcare first, then leverage that proven authority to expand across all markets.

Continuous Revenue

Transition from one-time audits to continuous automated monitoring subscriptions, locking in scalable and predictable monthly recurring revenue.

Agency Partnerships

White-label our tool for web design agencies, turning them into high-volume resellers who upgrade their existing client portfolios.

FUTURE SCOPE, ROADMAP & REFERENCES

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01

BROWSER EXTENSION & REAL-TIME DETECTION

A lightweight browser extension that detects and flags dark patterns instantly as users navigate, enabling proactive prevention instead of post-audit analysis.

02

MULTILINGUAL & CULTURAL COMPLIANCE

Models trained on diverse datasets to detect culturally nuanced dark patterns and ensure compliance with regional regulations like DPDP, LGPD, and PIPEDA.

03

DESIGNER CO-PILOT INTEGRATION

Integration with **design tools like Figma** to provide real-time dark pattern warnings during wireframing, preventing issues before development begins.

Build Core Detection Engine

Develop the fundamental detection engine with a precision target

Add Verification + Compliance Scoring

Integrate verification and compliance scoring mechanisms

Launch MVP

Release the Minimum Viable Product with a web interface and beta users

Subscription Model + Integrations

Establish a subscription model and integrate with Jira and API

Scale System + Fine-tune Models

Expand the system, refine models, and launch an extension

References:

- EU AI Act 2024
- FTC Dark Patterns Report
- Dark Patterns at Scale dataset
- Ultralytics YOLO-v26
- Distilbert

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